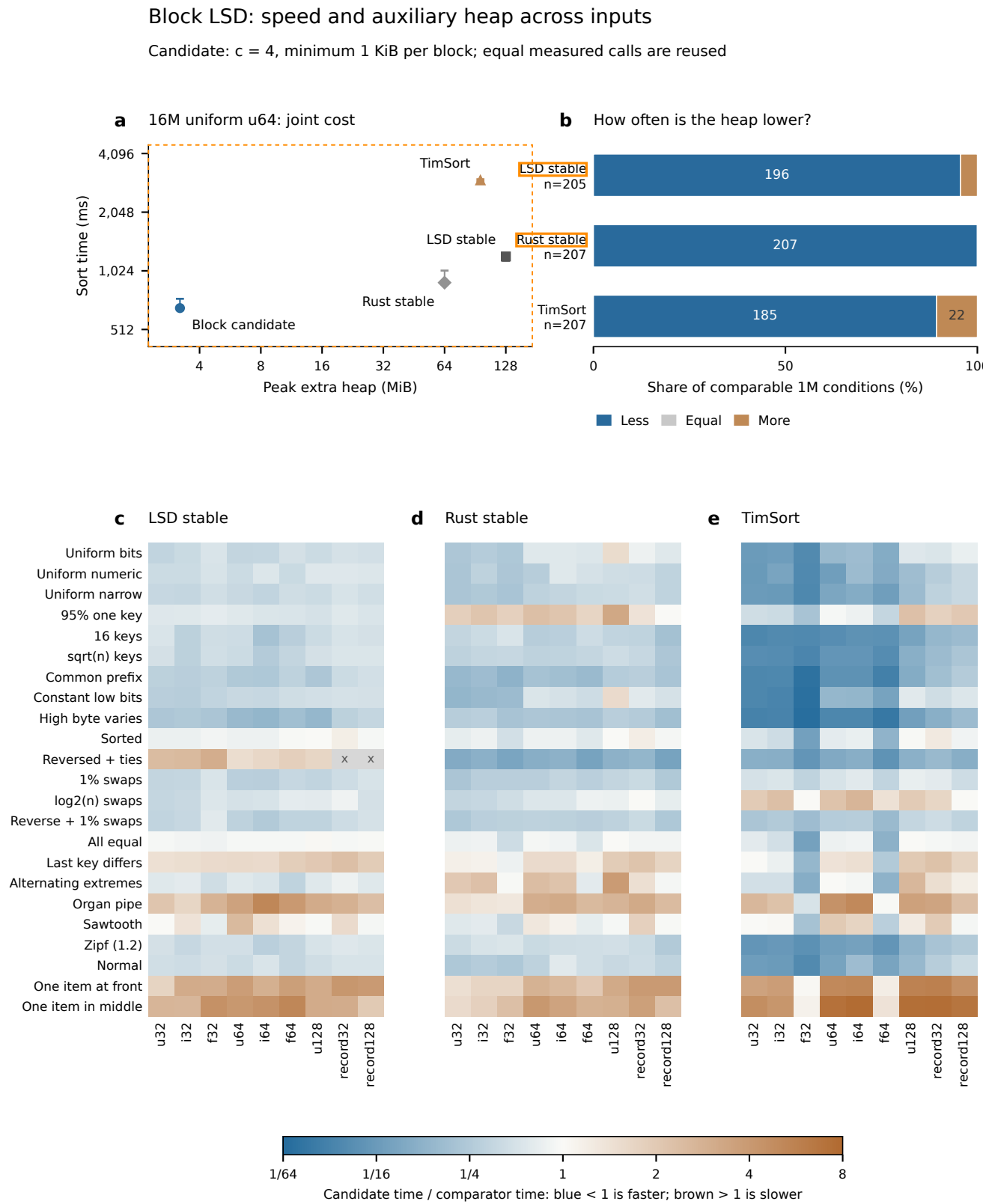
Block LSD: speed and auxiliary heap across inputs
Candidate: c = 4, minimum 1 KiB per block; equal measured calls are reused
a
16M uniform u64: joint cost
Sort time (ms)
4,096
2,048
1,024
512
TimSort
LSD stable
LSD stable
Rust stable
Rust stable
Block candidate
4
8
16
32
64
128
Peak extra heap (MiB)
b
How often is the heap lower?
LSD stable n=205
Rust stable n=207
TimSort n=207
196
207
185
22
0
50
100
Share of comparable 1M conditions (%)
Less
Equal
More
c
LSD stable
d
Rust stable
e
TimSort
Uniform bits
Uniform numeric
Uniform narrow
95% one key
16 keys
sqrt(n) keys
Common prefix
Constant low bits
High byte varies
Sorted
Reversed + ties
1% swaps
log2(n) swaps
Reverse + 1% swaps
All equal
Last key differs
Alternating extremes
Organ pipe
Sawtooth
Zipf (1.2)
Normal
One item at front
One item in middle
x
x
u32
i32
f32
u64
i64
f64
u128
record32
record128
1/64
1/16
1/4
1
2
4
8
Candidate time / comparator time: blue < 1 is faster; brown > 1 is slower
c-e: each cell is one type and distribution at 1M; median of six within-campaign paired time ratios.
Grey x: stable-output oracle failure (LSD, two reversed-record cases). a: medians and observed timing ranges.